鏡射與無鏡射的Aggregate

ONTAP 9

NetApp
May 03, 2022

This PDF was generated from https://docs.netapp.com/zh-tw/ontap/disks-aggregates/mirrored-unmirrored-aggregates-concept.html on May 03, 2022. Always check docs.netapp.com for the latest.

目錄

鏡射與無鏡射的Aggregate 1
鏡射與無鏡射的Aggregate 1
無鏡射Aggregate的運作方式 1
鏡射Aggregate的運作方式 2

鏡射與無鏡射的Aggregate

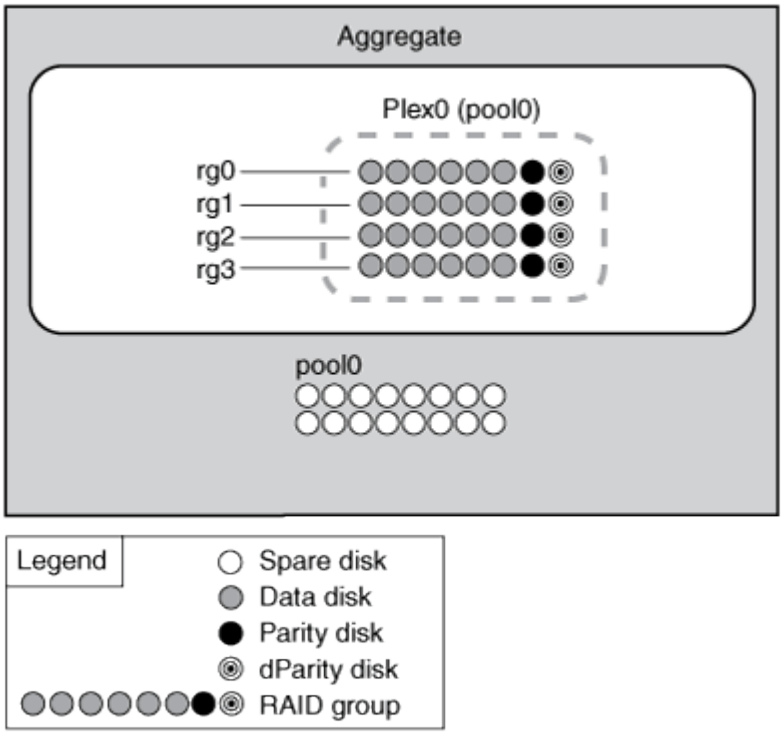
鏡射與無鏡射的Aggregate

您可以使用稱為「*SIRS* SyncMirror 」的選用功能、在儲存在不同RAID群組的複本或_plex_中同步鏡射集合體資料。如果有更多磁碟故障、或是RAID群組磁碟失去連線、則會利用叢碼來確保資料不會遺失。

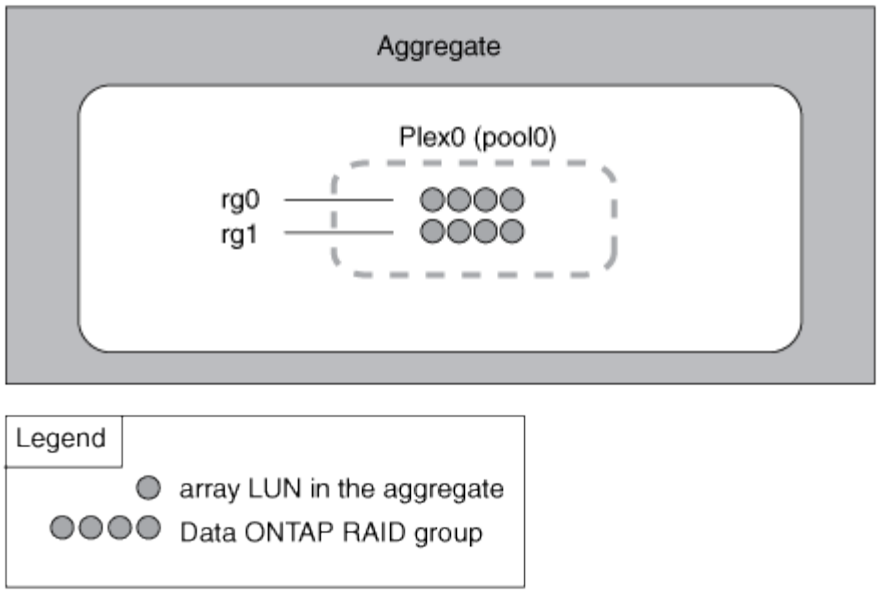
無鏡射Aggregate的運作方式

除非您使用SyncMirror 的是無法鏡射的全部集合體、無鏡射的Aggregate只有一個_plex（資料複本）、其中包含屬於該Aggregate的所有RAID群組。

下圖顯示由磁碟組成的無鏡射Aggregate、以及其單一叢。此Aggregate有四個RAID群組：rg0、rg1、rg2和rg3。每個RAID群組都有6個資料磁碟、一個同位元檢查磁碟和一個同位元檢查（雙同位元檢查）磁碟。集合體使用的所有磁碟都來自相同的集區Pool0。



下圖顯示陣列LUN與其單一叢的無鏡射Aggregate。它有兩個RAID群組：rg0和rg1。集合體使用的所有陣列LUN均來自相同的集區Pool0。



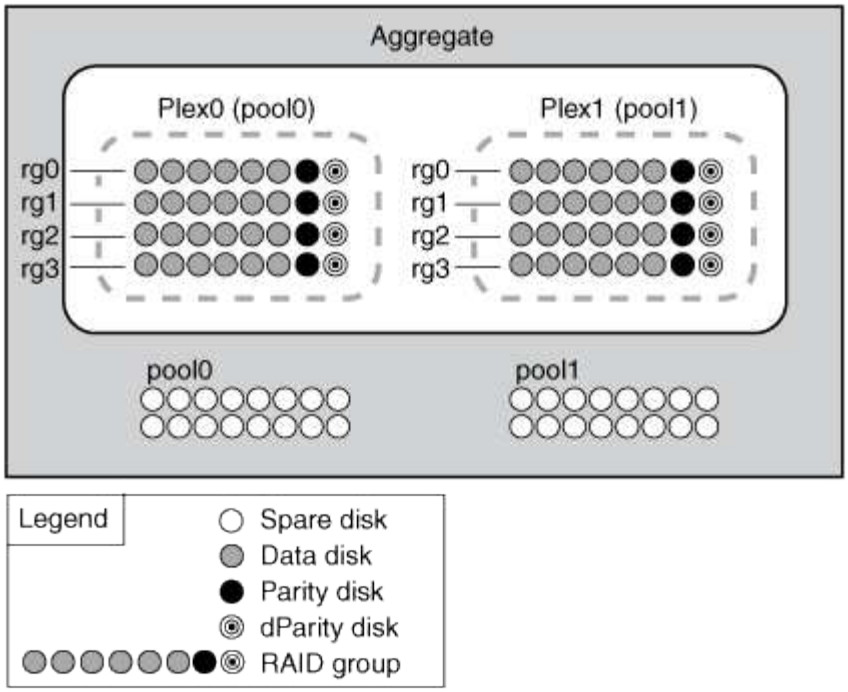
鏡射**Aggregate**的運作方式

鏡射Aggregate有兩個_plex_（資料複本）、這些複本使用SyncMirror 「畫面」功能來複製資料、以提供備援。

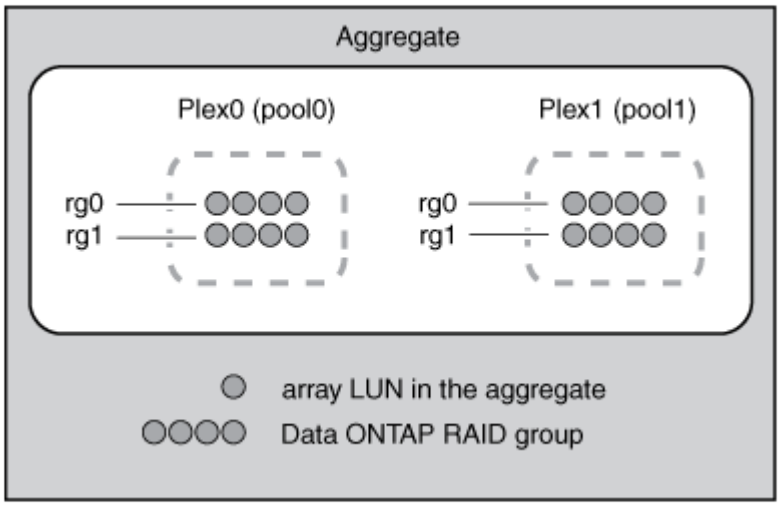
建立鏡射Aggregate（或將第二個叢新增至現有的無鏡射Aggregate）時ONTAP 、將原始叢（plex0）中的資料複製到新叢（plex1）。實體分離的叢集（每個叢集都有自己的RAID群組及其本身的資源池）、並同時更新叢集。如果有更多磁碟故障、而非集合體的RAID層級、則可提供額外的資料遺失保護功能、避免發生連線中斷、因為未受影響的叢會在您修正故障原因時、繼續提供資料。修復有問題的叢之後、兩個叢會重新同步、並重新建立鏡射關係。

系統上的磁碟和陣列LUN分為兩個集區：Pool0和Pool1。Plex0從Pool0取得其儲存設備、而plex1則從Pool1取得其儲存設備。

下圖顯示已SyncMirror 啟用及實作「支援功能」的磁碟集合體。已為Aggregate plex1建立第二個叢。plex1中的資料是plex0中的資料複本、RAID群組也相同。32個備用磁碟會配置給Pool0或Pool1、每個集區有16個磁碟。



下圖顯示由陣列LUN組成的集合體、SyncMirror 其中啟用並實作了各種功能。已為Aggregate plex1建立第二個叢。Plex1是plex0的複本、RAID群組也相同。



Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.